

Student Name \_\_\_\_\_ Student Class \_\_\_\_\_



2020

HSC Open Book Test Term 1

# Mathematics Extension 1

## *General Instructions*

- Reading time – 5 minutes
- Working time – 90 minutes
- Board approved calculators may be used.
- Write using black or blue pen.
- All necessary working should be shown in all questions.
- Write your student number at the top of every page of your answer sheets.
- You may bring in 1 page A4 paper with anything written or typed on it.

**Total marks – 61**

**Attempt all questions**

**Start each question on a new sheet of paper**

**A Formula Sheet is provided**

**This paper MUST NOT be removed from the examination room**

## Section A Multiple Choice (5 Marks)

### QUESTION 1

We can express  $\sin x$  and  $\cos x$  in terms of  $\tan \frac{x}{2}$ , for all values of  $x$  except.....

(A)  $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$ .....

(B)  $x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$ .....

(C)  $x = \pi, 3\pi, 5\pi$ .....

(D)  $x = 2\pi, 6\pi, 8\pi$ .....

### QUESTION 2

What is the value of  $k$  if  $\int_0^k \frac{dx}{9+x^2} = \frac{\pi}{9}$

(A)  $\sqrt{3}$

(B) 3

(C)  $\frac{1}{\sqrt{3}}$

(D)  $3\sqrt{3}$

### QUESTION 3

What is the coefficient of  $x^3$  in the expansion  $(1 - 3x)^7$

(A) -945

(B) -35

(C) 35

(D) 945

#### QUESTION 4

An advertisement claims that '8 out of 10 prefer Winky Chocolate Bars'. If the advertisement's claim is accurate and a sample of six people are interviewed, what is the probability that at least five people prefer Winky Chocolate Bars?

- (A)  $1 - (0.8)^4$
- (B)  $2(0.8)^5$
- (C)  $5(0.2)^5$
- (D)  $(0.8)^5(0.2) + (0.8)^6$

#### QUESTION 5

What is the derivative of  $y = \cos^{-1}(\frac{1}{x})$  with respect to  $x$ ?

- (A)  $\frac{-1}{\sqrt{x^2 - 1}}$
- (B)  $\frac{-1}{x\sqrt{x^2 - 1}}$
- (C)  $\frac{1}{\sqrt{x^2 - 1}}$
- (D)  $\frac{1}{x\sqrt{x^2 - 1}}$

**Section B Answer on the lined paper provided****Marks****QUESTION 6 (8 Marks) Start a new page**

(a) Evaluate  $\int_0^1 \frac{1}{\sqrt{4-2x^2}} dx$  leaving your answer in exact form. **3**

(b) Show that  $\int_0^{\frac{\pi}{16}} \cos^2 4x \, dx$  is  $\frac{1}{16} + \frac{\pi}{32}$  **2**

(c) Solve the equation  $\sin 2x = \tan x$  for  $0 \leq x \leq \pi$  **3**

**QUESTION 7 (8 Marks) Start a new page**

(a) Extensive research has shown that 1 person out of every 4 is allergic to a particular grass seed. A group of university students volunteer to try out a new treatment

(i) If 20 students are tested, what is the expected number of allergic people in the group? **1**

(ii) Find how large a sample would need to be for the probability of it containing at least one allergic person, to be greater than 99.9%? **3**

(iii) What assumptions have you made in your answer? **1**

(b) Evaluate  $\int_1^{49} \frac{1}{4(x+\sqrt{x})} dx$  using the substitution  $u^2 = x$ ,  $u > 0$  **3**

Give the answer in exact form.

**QUESTION 8 ( 8 Marks) Start a new page**

(a) Use the fact that for a restricted domain,  $y = \sin^{-1} x$  is equivalent to

2

$x = \sin y$  to show that  $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

(b) A spherical weather balloon is being inflated from empty using a container of helium, so that its volume is increasing at a constant rate of  $0.5 \text{ m}^3/\text{s}$ . (Assume the balloon maintains a spherical shape throughout inflation.)

(i) Show that the radius at a time  $t$  is given by  $r = \sqrt[3]{\frac{3t}{8\pi}}$

2

(ii) Show that the rate of increase of its surface area after 8 seconds is  $\sqrt[3]{\frac{\pi}{3}} \text{ m}^2/\text{s}$ .

3

(iii) If the maximum safe surface area before there is a risk that the balloon will burst is  $200 \text{ m}^2$ , what is the maximum time that the inflation should be allowed to proceed?

1

**QUESTION 9 (8 Marks) Start a new page**

(a)  $X_1, X_2, \dots, X_n$  are independent  $B(1, p)$  random variables.

Let  $Y = X_1 + X_2 + \dots + X_n$  and  $\hat{p} = \frac{Y}{n}$

(i) Given  $\text{var}(Y) = np(1-p)$  show  $\text{var}(\hat{p}) = \frac{p(1-p)}{n}$

2

(ii) Show  $0 \leq \text{var}(\hat{p}) \leq \frac{1}{4n}$

2

(b) Let  $X$  be a Bernoulli random variable with parameter  $p$ , where  $0 < p < 1$

(i) Explain why  $p$  cannot take the values  $p = 0$  or  $p = 1$ .

1

(ii) How many distinct values can  $X$  take in  $n$  Bernoulli trials?

1

(iii) A biased coin is tossed where  $P(\text{Head}) = \frac{1}{5}$ . If you obtain Heads you win a prize of \$1.

Could this situation be modelled as a Bernoulli random variable? Explain

2

**QUESTION 10 (8 Marks) Start a new page**

- (a) Use the substitution  $t = \tan \frac{\theta}{2}$  to prove the identity

$$\frac{1}{2}(3\cos\theta + 4\sin\theta + 5) = \left(\sin\frac{\theta}{2} + 2\cos\frac{\theta}{2}\right)^2 \quad 3$$

- (b) Show that  $\binom{n+2}{3} - \binom{n}{3} = n^2$  for all integers  $n$ , where  $n \geq 3$ . 2

- (c) Suppose that the true, unknown proportion  $p$  of voters who will vote for a third-party candidate in the next election is 9%. What is the probability that a poll of 820 voters will find a sample proportion  $\hat{p}$  that differs from the true proportion by more than 3%? You may use the normal distribution on the Reference sheet to approximate your answer to the nearest 0.1% 3

**QUESTION 11 (8 Marks) Start a new page**

- (a) The blades of a wind turbine are turning at a steady rate. The height,  $h$  metres, of the tip of one of the blades above the ground at time,  $t$  seconds, is given by the formula

$$h = 36\sin(1.5t) - 15\cos(1.5t) + 65$$

- (i) Express  $36\sin(1.5t) - 15\cos(1.5t)$  in the form  $k\sin(1.5t - a)$ , where  $k > 0$  and  $0 < a < 2\pi$ , 3

- (ii) Hence find the two values of  $t$  for which the tip of this blade is at a height of 100 metres above the ground during the first turn. 3

- (b) For the expansion  $(1 - 2x - 4x^2)\left(1 - \frac{3}{x}\right)^9$ , find the coefficient of the term in  $x^{-5}$  2

**QUESTION 12 (8 Marks) Start a new page**

In the expansion of  $(0.15 + 0.85)^{50}$ , the terms involving  $(0.15)^r$  and  $(0.15)^{r+1}$  are denoted by  $T_r$  and  $T_{r+1}$

(i) Show that  $\frac{T_r}{T_{r+1}} = \frac{17(r+1)}{3(50-r)}$  4

The probability that Paul's train to work is late on any day is 0.15, independently of other days. The number of days on which Paul's train is late during a 50-day period is modelled by the random variable  $X$

(ii) Find the values of  $r$  for which  $P(X = r) \leq P(X = r + 1)$  3

(iii) Hence find the most likely number of days on which the train will be late during the 50-day period. 1

**END OF EXAMINATION**



Student Number \_\_\_\_\_

Question 6

$$\begin{aligned} \text{(a)} \int_0^1 \frac{1}{\sqrt{4-2x^2}} dx &= \frac{1}{\sqrt{2}} \int_0^1 \frac{1}{\sqrt{2-x^2}} dx \quad \checkmark \quad \sqrt{4-2x^2} = \sqrt{2(2-x^2)} \\ &= \frac{1}{\sqrt{2}} \left[ \sin^{-1} \left( \frac{x}{\sqrt{2}} \right) \right]_0^1 \\ &= \frac{1}{\sqrt{2}} \left( \left[ \sin^{-1} \left( \frac{1}{\sqrt{2}} \right) \right] - \sin^{-1} \left( \frac{0}{\sqrt{2}} \right) \right) \checkmark \\ &= \frac{1}{\sqrt{2}} \left[ \frac{\pi}{4} - 0 \right] \\ &= \frac{\pi}{4\sqrt{2}} \checkmark \\ &= \frac{\sqrt{2}\pi}{8} \end{aligned}$$

$$\begin{aligned} \text{(b)} \int_0^{\frac{\pi}{16}} \cos^2 4x dx &= \frac{1}{2} \int_0^{\frac{\pi}{16}} \cos 8x + 1 dx \quad \checkmark \\ &= \frac{1}{2} \left[ \frac{\sin 8x}{8} + x \right]_0^{\frac{\pi}{16}} \\ &= \frac{1}{2} \left[ \frac{\sin \frac{\pi}{2}}{8} + \frac{\pi}{16} - \left( \frac{\sin 0}{8} + 0 \right) \right] \checkmark \\ &= \frac{1}{2} \left( \frac{1}{8} + \frac{\pi}{16} \right) \\ &= \frac{1}{16} + \frac{\pi}{32} \end{aligned}$$

$$\text{(c)} \sin 2x = \tan x \cdot 0 \leq x \leq \pi$$

$$2 \sin x \cos x = \frac{\sin x}{\cos x}$$

$$2 \sin x \cos^2 x = \sin x \quad \checkmark$$

$$\sin x (2 \cos^2 x - 1) = 0$$

$$\therefore \sin x = 0 \quad \text{OR} \quad 2 \cos^2 x - 1 = 0 \quad \checkmark$$

$$x = 0, \pi$$

$$\cos^2 x = \frac{1}{2}$$

$$\cos x = \pm \sqrt{\frac{1}{2}}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$\therefore x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi \quad \checkmark$$

# MATHEMATICS Extension 1: Question 7

## Suggested Solutions

Marks

Marker's Comments

(a) Expected number of allergic people

$$(i) E(X) = 0.25 \times 20 = 5$$

$\therefore 5$  allergic people in group ✓

(ii)  $P(\text{at least 1 allergic person}) = 1 - (0.75)^n$

$$1 - (0.75)^n > 0.999$$

$$(0.75)^n < 0.001$$

$$n \ln(0.75) < \ln 0.001$$

$$\therefore n > \frac{\ln 0.001}{\ln 0.75}$$

$$n > 24.61$$

$\therefore 25$  people are required.

(iii) accepted reasons: assumptions

- Random
- Independent
- Unbiased
- sample represents population
- probability of being allergic remains constant for all people
- Uni students have the same chance of being allergic as all other people

1

3

1 mark for recognising  $1 - 0.75^n > 0.999$   
1 mark solving for n - looking at steps  
1 mark for getting  $n = 24.61$  and rounding it up to get 25

1

MATHEMATICS Extension 1: Question 7

Suggested Solutions

Marks

Marker's Comments

$$7(b) \int_1^{49} \frac{1}{4(x+\sqrt{x})} dx$$

let  $u^2 = x$   
 $\frac{dx}{du} = 2u$   
 $\therefore dx = 2u \cdot du$

when  $x = 1$   
 $u = 1$   
 when  $x = 49$   
 $u = 7$

$$\int_1^7 \frac{1}{4(u^2+u)} \cdot 2u \cdot du$$

$$= \frac{1}{2} \int_1^7 \frac{u}{u^2+u} \cdot du$$

$$= \frac{1}{2} \int_1^7 \frac{1}{u+1} \cdot du$$

$$= \frac{1}{2} [\ln(u+1)]_1^7$$

$$= \frac{1}{2} [\ln 8 - \ln 2]$$

$$= \frac{1}{2} [\ln 4]$$

$$= \ln 2$$

3

✓ 1 mark for substituting  
 $* dx = 2u \cdot du$   
 $* x = u^2$   
 $* \sqrt{x} = u$

✓ 1 mark for integrating with correct integral.  
 ie  $\frac{1}{2} \int_1^7 \frac{1}{u+1} \cdot du$

Note this is not equal to  
 $\frac{1}{2} \int_1^7 \frac{1}{u} + 1$

✓ 1 substituting with correct limits  $u=1$  and  $u=7$   
 accepted  
 $\frac{1}{2} [\ln 8 - \ln 2]$

Question 8b)

Monday, 20 April 2020 12:18 pm

(i)  $\frac{dV}{dt} = 0.5$

$\therefore V = 0.5t + C$  when  $t=0, V=0$

$\therefore C=0$

$V = 0.5t$

$V = \frac{4}{3}\pi r^3$

$0.5t = \frac{4}{3}\pi r^3$

$t = \frac{8}{3}\pi r^3$

$r^3 = \frac{3t}{8\pi}$

$\therefore r = \sqrt[3]{\frac{3t}{8\pi}}$

①  $V = 0.5t$  after showing  $C=0$

① expression for  $t$

ii) Method 1: Simplest

$S = 4\pi r^2$

$\frac{dS}{dr} = 8\pi r$   $\frac{dt}{dr} = 8\pi r^2$

$\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt}$   
 $= 8\pi r \times \frac{1}{8\pi r^2}$

$= \frac{1}{r}$

$= \sqrt[3]{\frac{8\pi}{3t}}$   $r = \sqrt[3]{\frac{3t}{8\pi}}$

when  $t=8$

$\frac{dS}{dt} = \sqrt[3]{\frac{8\pi}{3 \times 8}}$   
 $= \sqrt[3]{\frac{\pi}{3}}$

Method 2:

$S = 4\pi r^2$ ,  $r = \sqrt[3]{\frac{3t}{8\pi}} = \sqrt[3]{\frac{3}{8\pi}} \times t^{1/3}$

$\frac{dS}{dr} = 8\pi r$

$\frac{dr}{dt} = \sqrt[3]{\frac{3}{8\pi}} \times \frac{1}{3} t^{-2/3}$   
 $= \sqrt[3]{\frac{3}{8\pi}} \times \frac{1}{3 \sqrt[3]{t^2}}$

$\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt}$   
 $= 8\pi r \times \sqrt[3]{\frac{3}{8\pi}} \times \frac{1}{3 \sqrt[3]{t^2}}$   
 $= 8\pi \times \sqrt[3]{\frac{3t}{8\pi}} \times \sqrt[3]{\frac{3}{8\pi}} \times \frac{1}{3 \sqrt[3]{t^2}}$

when  $t=8$

$\frac{dS}{dt} = 8\pi \times \sqrt[3]{\frac{3}{\pi}} \times \sqrt[3]{\frac{3}{8\pi}} \times \frac{1}{3 \sqrt[3]{8^2}}$   
 $= \frac{8\pi}{3} \times \sqrt[3]{\frac{9}{8^3 \pi^2}}$   
 $= \frac{9 \sqrt[3]{8^3 \times 9 \pi^3}}{3^3 8^3 \pi^2}$   
 $= \sqrt[3]{\frac{\pi}{3}}$

Marks:

① for  $\frac{dS}{dt} = \frac{1}{r}$

①  $\frac{dS}{dt}$  in terms of  $t$

① substitution of  $t=8$  and simplification.

(iii)  $4\pi r^2 = 200$

$r^2 = \frac{200}{4\pi}$

$= \frac{50}{\pi}$

$r = \sqrt{\frac{50}{\pi}}$ ,  $r > 0$

$\therefore \sqrt{\frac{50}{\pi}} = \sqrt[3]{\frac{3t}{8\pi}}$

$t = \left(\sqrt{\frac{50}{\pi}}\right)^3 \times \frac{8\pi}{3} = \frac{8 \times 50 \sqrt{50}}{3 \sqrt{\pi}} = \frac{400 \sqrt{50}}{3 \sqrt{\pi}}$

$= 531.9230405 \text{ s}$

$\therefore \text{max. time} = 531.9 \text{ s}$

or  $531.9 \text{ s}$

or  $8 \text{ min } 51.9 \text{ s}$

① no penalty for subsequent error of rounding up to 532s which is too long.

## Question 8a)

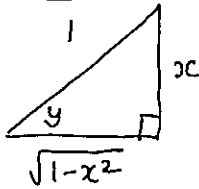
Monday, 20 April 2020

12:07 pm

a)  $y = \sin^{-1} x$

$x = \sin y$

$\frac{dx}{dy} = \cos y$

Method 1:

$\cos y = \sqrt{1-x^2}$

$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

Method 2:

$\cos^2 y + \sin^2 y = 1$

$\cos^2 y = 1 - \sin^2 y$

$\cos y = \sqrt{1 - \sin^2 y}$

since for  $y = \sin^{-1} x$ ,  $-\pi/2 \leq y \leq \pi/2$ and  $\cos x > 0$  for this domain

$\therefore \frac{dx}{dy} = \sqrt{1 - \sin^2 y}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1 - \sin^2 y}}$

$= \frac{1}{\sqrt{1-x^2}}$

① for finding  $\frac{dx}{dy} = \cos y$ Method 1:① for using triangle to show  $\cos y = \sqrt{1-x^2}$ 

and  $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

note that  $\frac{dy}{dx} \neq \frac{dx}{dy}$ 

① for using Pythagorean identity to

show  $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks             | Marker's Comments |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|
| <p>a) i) <math>x_1, x_2, \dots, x_n, B(1, p)</math><br/> <math>y = x_1 + x_2 + \dots + x_n, \hat{p} = \frac{y}{n}</math> } given.<br/> <math>Var(Y) = np(1-p)</math></p> <p>Show <math>var(\hat{p}) = \frac{p(1-p)}{n}</math></p> <p>we know <math>Var(Y) = npq, q = (1-p)</math></p> <p><u>M1</u></p> $\begin{aligned} var(\hat{p}) &= var\left(\frac{Y}{n}\right) \\ &= E\left(\left(\frac{Y}{n}\right)^2\right) - E\left(\frac{Y}{n}\right)^2 \\ &= \frac{1}{n^2} E(Y^2) - \frac{1}{n^2} [E(Y)]^2 \\ &= \frac{1}{n^2} [E(Y^2) - (E(Y))^2] \\ &= \frac{1}{n^2} Var(Y) \\ &= \frac{1}{n^2} (np(1-p)) \text{ from above} \\ &= \frac{p(1-p)}{n} \text{ shown.} \end{aligned}$ <p><u>M2</u></p> <p>Given <math>Var(Y) = np(1-p)</math><br/> <math>Var(\hat{p}) = var\left(\frac{Y}{n}\right)</math><br/> <math>= \frac{1}{n^2} Var(Y) \rightarrow</math> (since variance is the square of std dev)<br/> <math>= \frac{1}{n^2} (np(1-p))</math><br/> <math>= \frac{p(1-p)}{n}</math> (shown) example from maths in Fom 12 (Ext 1) pg 692.</p> | <p>1</p> <p>1</p> |                   |

Suggested Solutions

Marks

Marker's Comments

a) Show  $0 \leq \text{var}(\hat{p}) \leq \frac{1}{4n}$

$$y = p(1-p)$$

$$\frac{dy}{dp} = 1 - 2p$$

$$\text{when } \frac{dy}{dp} = 0$$

$$p = \frac{1}{2}$$

|                 |   |               |    |
|-----------------|---|---------------|----|
| $p$             | 0 | $\frac{1}{2}$ | 1  |
| $\frac{dy}{dp}$ | 1 | 0             | -1 |
| shape/sign      | / | -             | \  |

$$y_{\text{max}} \text{ at } p = \frac{1}{2}$$

$$= \frac{1}{2} \left(1 - \frac{1}{2}\right)$$

$$= \frac{1}{4}$$

- greatest value of  $p(1-p)$  occurs when  $p = \frac{1}{2}$ ;  $p(1-p) = \frac{1}{4}$

- Smallest value of  $p(1-p)$  occurs when  $p = 0$  or  $1$ ;  $p(1-p) = 0$

- smallest value of  $\text{var}(\hat{p}) = \frac{0}{n} = 0$

- greatest value of  $\text{var}(\hat{p}) = \frac{1}{4} = \frac{1}{4n}$

$$\therefore 0 \leq \text{var}(\hat{p}) \leq \frac{1}{4n}$$

b) i) - If  $p = 0$  or  $p = 1$ , there is a 100% chance that the trial will be a failure or success.

- A Bernoulli trial is an experiment that has 2 outcomes, either failure or success.

- If there is a 100% chance of either occurring, that means that there is only one outcome.

$\therefore$  The experiment will not be a Bernoulli trial becoz  $p$  cannot be 0 or 1

① - mark to show understanding of stationary points and concave down

① - mark for stating max tp at  $p = \frac{1}{2}$  and  $y_{\text{max}}$  at  $\frac{1}{4}$

or have shown a graphical interpretation.

② For why  $\text{var}(\hat{p}) \geq 0$ .

① - mark.

Have to mention something along the line why/how Bernoulli trial happens.

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks Awarded                                                             | Marker's Comments                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>b) ii) How many distinct values can <math>X</math> take in 'n' Bernoulli trials?</p> <p>The answer is <u>2</u> or <u>0 or 1</u></p>                                                                                                                                                                                                                                                                                                                                                                 | 1                                                                         | <p>Many students interpreted it incorrectly.</p> <p>The question said <math>X = \text{Bernoulli random variable}</math></p> <p><u>NOT</u></p> <p>Binomial random variable - This was the issue - for Bernoulli random variable - there are only 2 distinct values <math>X</math> can take. i.e. 2 or 0 or 1</p> |
| <p>b) iii) <math>P(H) = \frac{1}{5}</math> if heads win \$1</p> <p>Is this Bernoulli random variable, (BRV)</p> <p>ie from my comment, BRV has 2 outcomes; if <math>P(H) = \frac{1}{5}</math> <math>P(\bar{H}) = \frac{4}{5}</math></p> <p style="text-align: center;">win                      lose.<br/>(1)                      (0)</p> <p style="text-align: center;">as per definition.</p> <p>Therefore the answer is yes and the second mark is awarded if student explains this correctly.</p> | <p style="text-align: center;">1</p> <p style="text-align: center;">1</p> | <p style="text-align: center;">1</p> <p style="text-align: center;">1</p>                                                                                                                                                                                                                                       |

Q10.

a) Let  $t = \tan \frac{\theta}{2}$  (Method 1)

$$\therefore \frac{1}{2} (3 \cos \theta + 4 \sin \theta + 5)$$

$$= \frac{1}{2} \left( 3 \times \frac{1-t^2}{1+t^2} + 4 \times \frac{2t}{1+t^2} + 5 \right)$$

$$= \frac{1}{2} \left( \frac{3-3t^2}{1+t^2} + \frac{8t}{1+t^2} + \frac{5+5t^2}{1+t^2} \right)$$

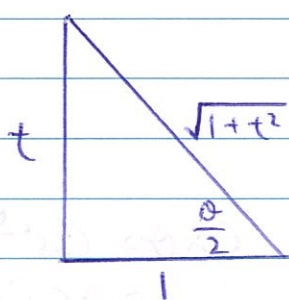
$$= \frac{1}{2} \left( \frac{3-3t^2+8t+5+5t^2}{1+t^2} \right)$$

$$= \frac{1}{2} \left( \frac{2t^2+8t+8}{1+t^2} \right)$$

$$= \frac{t^2+4t+4}{1+t^2}$$

$$= \frac{(t+2)^2}{1+t^2}$$

① Arriving at this expression from LHS



$$\sin \frac{\theta}{2} = \frac{t}{\sqrt{1+t^2}}$$

$$\cos \frac{\theta}{2} = \frac{1}{\sqrt{1+t^2}}$$

① This step is a must (including the triangle) as students are not allowed to quote  $\sin \frac{\theta}{2}$  and  $\cos \frac{\theta}{2}$  in a "prove" question

$$\therefore \text{RHS} = \left( \frac{t}{\sqrt{1+t^2}} + 2 \times \frac{1}{\sqrt{1+t^2}} \right)^2$$

$$= \left( \frac{t+2}{\sqrt{1+t^2}} \right)^2$$

$$= \frac{(t+2)^2}{1+t^2}$$

$$= \text{LHS}$$

① Arriving at this expression from RHS.

or After arriving LHS =  $\frac{(t+2)^2}{1+t^2}$  like in method 1

(Method 2)

$$\text{LHS} = \frac{(t+2)^2}{1+t^2} \quad \text{--- (1)}$$

$$= \frac{\left(\tan \frac{\theta}{2} + 2\right)^2}{1 + \tan^2 \frac{\theta}{2}}$$

$$= \frac{\left(\frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} + 2\right)^2}{\sec^2 \frac{\theta}{2}}$$

Getting rid of  $t$  or  $\tan \frac{\theta}{2}$  in the expression

$$= \cos^2 \frac{\theta}{2} \left( \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} + 2 \right)^2$$

$$= \left[ \cos \frac{\theta}{2} \left( \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} + 2 \right) \right]^2$$

$$= \left( \sin \frac{\theta}{2} + 2 \cos \frac{\theta}{2} \right)^2$$

Arriving to RHS. (1)

What NOT to do

Intention of Qn

$$\text{RHS} = \left( \sin \frac{\theta}{2} + 2 \cos \frac{\theta}{2} \right)^2$$

$$= \sin^2 \frac{\theta}{2} + 4 \sin \frac{\theta}{2} \cos \frac{\theta}{2} + 4 \cos^2 \frac{\theta}{2}$$

$$= 1 + 3 \cos^2 \frac{\theta}{2} + 4 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$= 1 + \frac{3 \cos \theta + 3}{2} + 2 \sin \theta \quad *$$

$$\cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}$$

$$= 2 \cos^2 \frac{\theta}{2} - 1$$

$$\therefore \cos^2 \frac{\theta}{2} = \frac{\cos \theta + 1}{2}$$

$$\text{Sub } \cos \theta = \frac{1-t^2}{1+t^2} \text{ and } \sin \theta = \frac{2t}{1+t^2}$$

$$\therefore \text{RHS} = \frac{(t+2)^2}{1+t^2}$$

- Those who approached RHS this way gets 1/3 MAX from for the entire question as you've essentially done the prove up to \* without  $t$  formula.

$$b) \text{ LHS} = \frac{(n+2)!}{[(n+2)-3]! \cdot 3!} - \frac{n!}{(n-3)! \cdot 3!}$$

$$= \frac{(n+2)!}{(n-1)! \cdot 3!} - \frac{n!}{(n-3)! \cdot 3!} \quad \text{--- (1)}$$

$$= \frac{n(n+1)(n+2)}{3!} - \frac{n(n-1)(n-2)}{3!}$$

$$= \frac{n[(n^2+3n+2) - (n^2-3n+2)]}{6}$$

$$= \frac{n \times 6n}{6} \quad \text{--- (1)}$$

$$= n^2$$

$$c) p=0.09 \quad q=0.91, \quad n=820$$

$$\sigma = \sqrt{npq} \doteq 8.195 \quad \text{--- (1)}$$

using formula for S.D

$$3\% \text{ of } 820 = 24.6$$

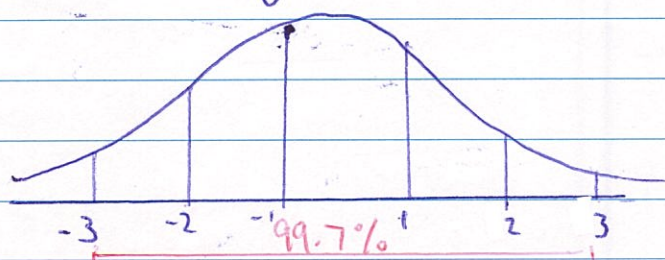
$$9\% \text{ of } 820 = 73.8$$

$\therefore$  looking for  $P(49.2 \leq X \leq 98.4)$

$$Z_{49.2} = \frac{49.2 - 73.8}{\sigma} \doteq -3.002 \quad \text{--- (1)}$$

one of the Z-scores

$$Z_{98.4} = \frac{98.4 - 73.8}{\sigma} \doteq 3.002$$



$$\therefore P(|X - 73.8| > 24.6) = 0.03\% \quad \text{--- (1)}$$

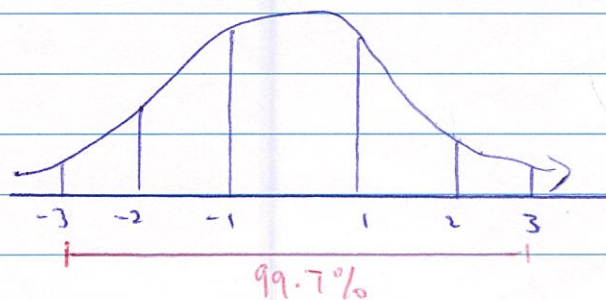
$$\sigma = \sqrt{\frac{pq}{n}} \doteq 9.994 \times 10^{-3} \quad \text{--- (1)}$$

using formula for S.D

$$P(0.06 \leq \hat{p} \leq 0.12) :$$

$$Z_{0.06} = \frac{0.06 - 0.09}{\sigma} \doteq -3.002 \quad \text{--- (1)}$$

$$Z_{0.12} = \frac{0.12 - 0.09}{\sigma} \doteq 3.002$$



$$\therefore P(|\hat{p} - 0.09| > 0.03) = 0.03\% \quad \text{--- (1)}$$

Q11 - marking standards

a) i)  $h = 36 \sin(1.5t) - 15 \cos(1.5t) + 65$

→ ①  $k = \sqrt{36^2 + 15^2} = \sqrt{1521} = 39$  either  $\sqrt{1521}$  or 39 - 1 mark.

→ ②  $\tan \alpha = \frac{15}{36} = \frac{5}{12} \Rightarrow \alpha = \tan^{-1}\left(\frac{5}{12}\right)$

or  $\sin \alpha = \frac{5}{13} \Rightarrow \alpha = \sin^{-1}\left(\frac{5}{13}\right)$  - either - 1 mark.

or  $\cos \alpha = \frac{12}{13} \Rightarrow \alpha = \cos^{-1}\left(\frac{12}{13}\right)$  -  $\alpha$  can have only one value.

or  $\alpha = 0.39479... \quad (\alpha \text{ can't be in degree form})$   
 $= 22^\circ 37' \text{ or } 22.62^\circ \text{ (x)}$

→ ③  $39 \sin(1.5t - \tan^{-1}(\frac{5}{12}))$  expression - 1 mark.

→ or any of the 4 angle forms above

ii)  $h = 39 \sin(1.5t - \tan^{-1}(\frac{5}{12})) + 65 = 100$

no mark if sin is missing from the expression

→ ①  $39 \sin(1.5t - \tan^{-1}(\frac{5}{12})) = 35$

$\sin(1.5t - \tan^{-1}(\frac{5}{12})) = \frac{35}{39}$   
 radian form → degree form (ECF)

correct sin ratio - 1 mark.

→ ②  $1.5t - \tan^{-1}(\frac{5}{12}) = \sin^{-1}(\frac{35}{39})$

$1.5t = \tan^{-1}(\frac{5}{12}) + \sin^{-1}(\frac{35}{39})$

$t = \frac{2}{3}(\tan^{-1}(\frac{5}{12}) + \sin^{-1}(\frac{35}{39}))$

$= 1.0058089...$

$\sin^{-1}(\frac{35}{39}) \div 63.49^\circ$

$t = 57^\circ 37' 42.99''$  correct

$= 57.6286^\circ$  t-value ① - 1 mark.

→ or ③  $1.5t - \tan^{-1}(\frac{5}{12}) = \pi - \sin^{-1}(\frac{35}{39})$

$t = \frac{2}{3}(\pi - \sin^{-1}(\frac{35}{39}) + \tan^{-1}(\frac{5}{12}))$

$= 1.614974...$

$t = 92^\circ 31' 52.36''$

$= 92.53121^\circ$  correct t-value ② - 1 mark.

$$b) (1 - 2x - 4x^2) \left(1 - \frac{3}{x}\right)^9$$

$$= (1 - 2x - 4x^2) \left( {}^9C_0 + {}^9C_1 \left(-\frac{3}{x}\right)^1 + {}^9C_2 \left(\frac{-3}{x}\right)^2 + \dots + {}^9C_5 \left(\frac{-3}{x}\right)^5 + {}^9C_6 \left(\frac{-3}{x}\right)^6 + {}^9C_7 \left(\frac{-3}{x}\right)^7 \right)$$

$$\text{Terms in } x^{-5} \quad {}^9C_5 (-3)^5 - 2 {}^9C_6 (-3)^6 - 4 {}^9C_7 (-3)^7$$

— 1 mark

$$= -30618 - 2 \times 61236 - 4 \times (-78732)$$

$$= -30618 - 122472 + 314928$$

$$= 161838$$

— 1 mark

or 3 correct expressions but wrong calculation

— 1 mark/2

or  $\begin{cases} 2 \text{ correct expressions} + \text{correct calculation} \\ 1 \text{ wrong expression} + \text{correct ECF calculation} \end{cases}$

— 1 mark/2

or in the expressions, forget to include "-" from "(-3)"

— 1 mark/2

or in the expressions, forget to consider "()" from "(-3)"

— 1 mark/2

$$\begin{aligned}
 \textcircled{12} \quad \frac{T_r}{T_{r+1}} &= \frac{\frac{50!}{r!(50-r)!} \times 0.15^r \times (0.85)^{50-r}}{\frac{50!}{(r+1)!(50-(r+1))!} \times 0.15^{r+1} \times 0.85^{50-(r+1)}} \\
 &= \frac{r+1}{50-r} \times \frac{0.85}{0.15} \\
 &= \frac{85(r+1)}{15(50-r)} \\
 &= \frac{17(r+1)}{3(50-r)}
 \end{aligned}$$

$$(ii) \quad P(x=r) \leq P(x=r+1)$$

$$\frac{P(x=r)}{P(x=r+1)} \leq 1 \quad \text{ie} \quad \frac{T_r}{T_{r+1}} \leq 1$$

$$\therefore \frac{17(r+1)}{3(50-r)} \leq 1$$

$$17r + 17 \leq 150 - 3r$$

$$20r \leq 133$$

$$r \leq 6.65$$

$\therefore$  values will be  $r=0, 1, 2, \dots, 6$ .

(iii) most likely number of days late is 7.